Attributes & Transformations of Quadratic Functions

Algebra II · Lesson 5.1 · Classical Notes

Name: _____ Date: _____ Period: _____

I. Definitions — copy each one exactly

Quadratic function

A function that can be written $f(x) = ax^2 + bx + c$, where $a \neq 0$.

Vertex form

$f(x) = a(x - h)^2 + k$. The vertex is visible directly: (h, k).

Vertex

The highest or lowest point of the parabola.

Axis of symmetry

The vertical line $x = h$, across which the parabola mirrors itself.

II. Demonstration — supply the reason for each step

Describe the transformations of $g(x) = -2(x - 3)^2 + 1$ compared to the parent function $y = x^2$. State the vertex and the axis of symmetry.

1. In vertex form $a(x - h)^2 + k$: $a = -2$, $h = 3$, $k = 1$.
   Reason: _____
2. Vertex: $(3, 1)$. Axis of symmetry: $x = 3$.
   Reason: _____
3. $|-2| = 2 > 1$, so the parabola is stretched vertically by a factor of 2 — narrower than the parent.
   Reason: _____
4. $a = -2 < 0$, so the parabola is reflected across the x-axis — it opens downward.
   Reason: _____
5. $h = 3$ shifts the graph right 3 units; $k = 1$ shifts it up 1 unit.
   Reason: _____

III. Questions for Discussion

1. Why does (x − 3) shift the graph right when 3 is being subtracted?

2. How does the single value a control both the width and the direction of the parabola?

3. What would change — and what would stay the same — if a were −½ instead of −2?

IV. Narration — explain it as if teaching a friend

How do you read the vertex and every transformation directly from vertex form, without graphing?

V. Error Log

My mistake	The correction

Q.E.D.